

PROBLEM STATEMENT #65 | Sustainability & Green Tech

Urban Rooftop Gardening Co-op App

1. Problem Context & Overview

A community urban agriculture app for registering rooftop plots, logging crop yields and harvest schedules, sharing soil maintenance tips, and trading surplus produce.

Target Stakeholders / Actors: *Co-op Gardener, Community Leader*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall allow gardeners to log seasonal crop planting dates, computing estimated harvest windows and expected yield weight.

Sample Acceptance Criteria: Pass: Harvest alert generated within predicted window; Fail: Invalid planting date sequence accepted.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The produce trade board must update available surplus listings in real-time across community members.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario >> One Alternate Flow.